

# Literature Status: The $\text{cov}(\mathcal{N})$ Lower Bound for Normed Barrelled Spaces

Automatic Math Discovery Run `fa_banach_001`

June 16, 2026

## Status

This is a **literature\_already\_answered** packet. It records that a later note explicitly answers Question 4.13 from arXiv:2410.13970. It is not a new proof packet.

## Source Question

The source paper is:

Will Brian and Christopher Stuart, *Small-dimensional normed barrelled spaces*, arXiv:2410.13970.

In Question 4.13, the authors ask:

Does every barrelled subspace of an infinite-dimensional Banach space have dimension  $\geq \text{cov}(\mathcal{N})$ ? In particular, is  $\text{cov}(\mathcal{N}) \leq \mathfrak{v}$ ?

Here  $\text{cov}(\mathcal{N})$  is the covering number of the Lebesgue null ideal, and  $\mathfrak{v}$  is the least algebraic dimension of an infinite-dimensional normed barrelled space.

## Supporting Answer

The supporting paper is:

Damian Sobota, *A small remark on small-dimensional normed barrelled spaces*, arXiv:2511.13355.

The note states Question 4.13 of arXiv:2410.13970 explicitly and says that its aim is to answer it affirmatively. Its main theorem, Theorem `thm:main`, proves that whenever  $A$  is a subset of an infinite-dimensional Banach space  $E$  and  $\text{span}(A)$  is barrelled, then

$$|A| \geq \text{cov}(\mathcal{N}).$$

The proof combines the Brian–Stuart method with Dvoretzky’s theorem, via the intermediate property  $(\dagger')$ , which Sobota proves is satisfied by every infinite-dimensional Banach space.

## Identification

Corollary `cor:covN` of arXiv:2511.13355 gives the source question in exact form:

every infinite-dimensional normed barrelled space has dimension at least  $\text{cov}(\mathcal{N})$ .

Corollary `cor:y_covN` then records the resulting estimate

$$\max(\mathfrak{b}, \text{cov}(\mathcal{N})) \leq \mathfrak{y} \leq \text{non}(\mathcal{M}).$$

Thus arXiv:2511.13355 gives a positive answer to Question 4.13 of arXiv:2410.13970.

## Scope Limitations

This packet clears only the  $\text{cov}(\mathcal{N})$ -lower-bound question. It does not settle the later residual question asked at the end of arXiv:2511.13355, namely whether  $\mathfrak{y} = \text{non}(\mathcal{M})$  holds in ZFC.

## Search Evidence

Local cheap indexes for arXiv:2410.13970, “small-dimensional normed barrelled”, “barrelled subspace”, and “ $\text{cov}(\mathcal{N})$ ” had no prior exact packet at reservation time. The run target pool listed arXiv:2511.13355, which directly cites Brian–Stuart’s Question 4.13 and answers it by Theorem `thm:main` and Corollary `cor:covN`.

## References

- [1] W. Brian and C. Stuart, *Small-dimensional normed barrelled spaces*, arXiv:2410.13970.
- [2] D. Sobota, *A small remark on small-dimensional normed barrelled spaces*, arXiv:2511.13355.